



Visualizing genus zero dessins

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Background

- Number theory, computation

Past

- Postdoc (ICERM)
- Mathematician (CRREL)
 - Computer vision, mixed models

Current

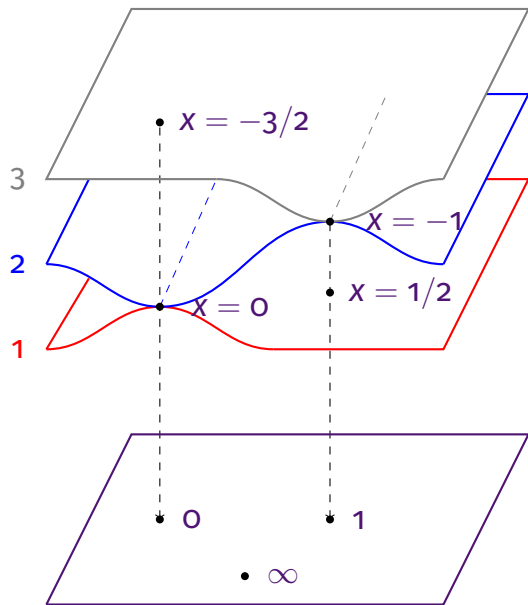
- Data wrangler (Faraday)
 - Product implementation
 - Modeling / API infrastructure
 - Client solutions
- Toddler wrangler

Outline



1. Background
2. Visualizations and objectives
3. Examples

A branched covering



$$\begin{aligned} \varphi(x) &= \underbrace{2x^3 + 3x^2}_z \\ &= x^2(2x + 3) \\ &= (2x - 1)(x + 1)^2 + 1 \end{aligned}$$

X

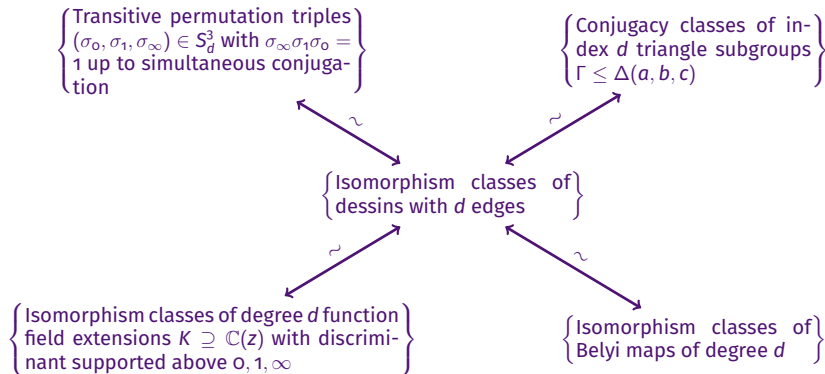
$\mathbb{P}^1$

Belyi maps, dessins, etc.



- This branched covering is an example of a **Belyi map**, a map of Riemann surfaces unbranched outside of $0, 1, \infty$.
- A Belyi map determines a **permutation triple** via monodromy.
- A permutation triple determines a **triangle subgroup** Γ which acts on a spherical, Euclidean, or hyperbolic space $\mathcal{H}$ such that $\Gamma \backslash \mathcal{H} \rightarrow \Delta \backslash \mathcal{H}$ is the corresponding Belyi map.
- The preimage of $[0, 1]$ is a (connected) bipartite graph embedded on a surface called a **dessin d'enfant**.

Equivalent categories

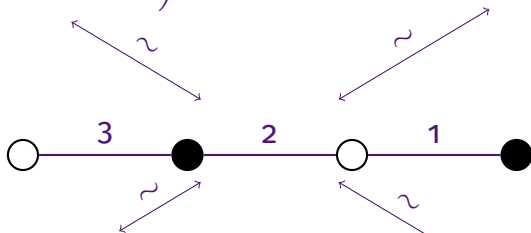


Degree 3 example



$$((12)(3), (1)(23), (132))$$

$$[\Delta(2, 2, 3) : \Gamma] = 3$$



$$\mathbb{C} \left(\underbrace{2x^3 + 3x^2}_z \right) \subseteq \mathbb{C}(x)$$

$$\varphi(x) = \underbrace{2x^3 + 3x^2}_z$$



Theorem (Belyi, 1979)

A smooth projective curve X over $\mathbb{C}$ can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a nonconstant morphism of algebraic curves $\varphi : X \rightarrow \mathbb{P}^1$ unramified outside $\{0, 1, \infty\}$.

In the 1980s Grothendieck described an action of the absolute Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on the set of (isomorphism classes of) dessins.

Passports



A **passport** consists of the data (g, G, λ) where

- $g \geq 0$ is an integer
- $G \leq S_d$ is a transitive subgroup
- $\lambda = (\lambda_0, \lambda_1, \lambda_\infty)$ is a triple of partitions of d

A permutation triple $\sigma \in S_d^3$ **belongs** to (g, G, λ) if

- $2g - 2 = -2d + \sum_{s \in \{0,1,\infty\}} \sum_{\tau \text{ a cycle in } \sigma_s} (\text{len}(\tau) - 1)$
- $\langle \sigma_0, \sigma_1, \sigma_\infty \rangle = G$
- $\sigma_0, \sigma_1, \sigma_\infty$ have the cycle types specified by $\lambda_0, \lambda_1, \lambda_\infty$

The **size** of a passport is the number of permutation triples belonging to it (up to simultaneous conjugacy).

Objective of this work



Draw topologically correct dessins to help visualize the Galois orbits of all (genus zero) passports in the LMFDB.

- <https://beta.lmfdb.org/Belyi/>
- <https://michaelmusty.github.io/dessins/>



- <https://beta.lmfdb.org/Belyi/7T6/5.1.1/4.2.1/3.2.2/>
- https://michaelmusty.github.io/dessins/passports/7T6-5.1.1_4.2.1_3.2.2/index.html
- All drawings together



- <https://beta.lmfdb.org/Belyi/5T4/5/5/3.1.1/>
- https://michaelmusty.github.io/dessins/passports/5T4-5_5_3.1.1/index.html
- All drawings together



- <https://beta.lmfdb.org/Belyi/9T34/6.1.1.1/5.2.2/6.1.1.1/>
- https://michaelmusty.github.io/dessins/passports/9T34-6.1.1.1_5.2.2_6.1.1.1/index.html
- All drawings together

Thanks for listening!